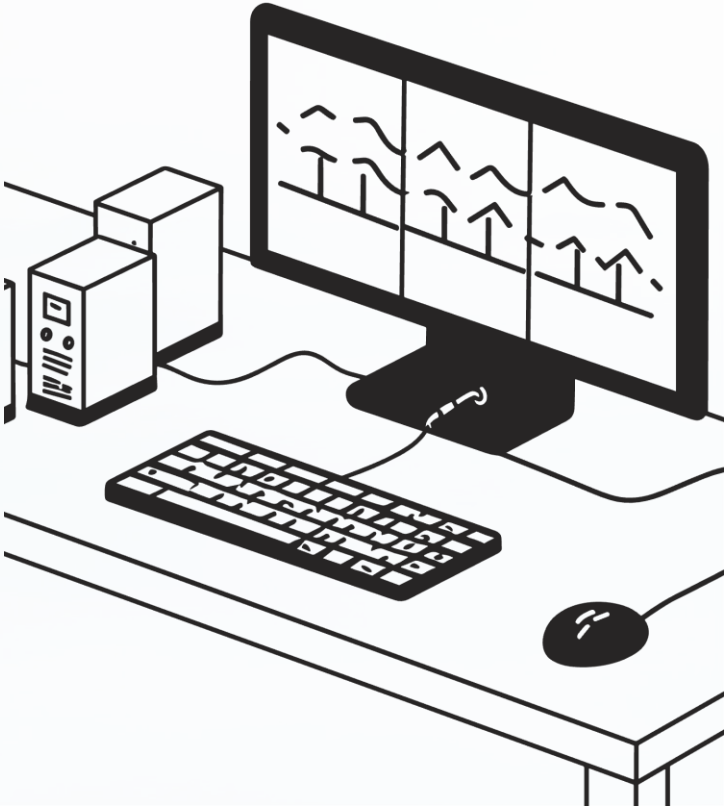
MACHINE LEARNING INTERNSHIP PROJECT · SUMMER 2026

Employee Attrition Prediction System

Developed by **Aditya Verma** · B.Tech CSE (AI & ML)
United College of Engineering & Research, Prayagraj · AKTU, Lucknow

Internship at **IIHMF**, MNNIT Prayagraj



INTRODUCTION

Problem Statement



What is Employee Attrition?

The voluntary or involuntary departure of employees from an organization — a costly challenge impacting productivity and morale.

Why Predict It?

- Replacement costs can exceed **1.5× annual salary**
- Disrupts team dynamics and project continuity
- ML enables **proactive, data-driven** retention strategies
- HR analytics transforms reactive hiring into **predictive workforce planning**

OBJECTIVES & DATASET

Project Goals & Data Source

Project Objectives

- Predict employee attrition using supervised ML models
- Compare Logistic Regression, Random Forest, and XGBoost
- Deploy a real-time prediction system via Streamlit

IBM HR Analytics Dataset

1,470

Employee records

35

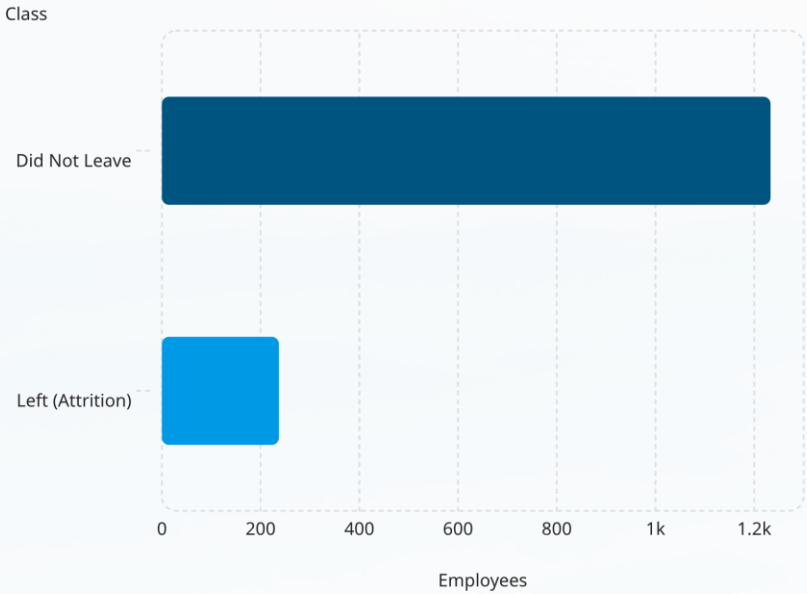
Features

Binary

Target: Attrition (Yes/No)

EXPLORATORY DATA ANALYSIS

Understanding the Data



Key Observations

- **Class Imbalance:** ~16% attrition rate — minority class requires careful handling
- **Correlation Analysis:** Overtime, Job Satisfaction, and Years at Company were top predictors
- **Categorical Features:** Department, Job Role, and Business Travel showed meaningful distribution patterns
- **No Missing Values** in the original dataset — clean and well-structured

DATA PREPROCESSING

Preparing the Data for Modeling



A clean, consistent preprocessing pipeline ensures all three models are trained on identical, well-structured input data — enabling fair and reproducible comparison.

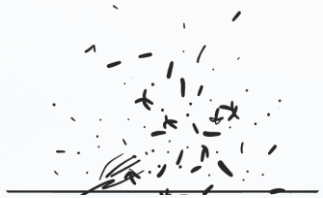
MACHINE LEARNING MODELS

Three Models, One Dataset



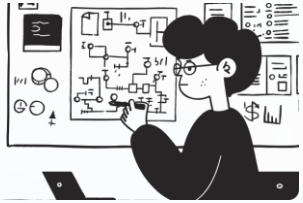
Logistic Regression

Linear probabilistic classifier. Interpretable, efficient, and well-suited for binary classification tasks.



Random Forest

Ensemble of decision trees. Robust to overfitting, handles non-linear relationships naturally.



XGBoost

Gradient boosting framework. High predictive power with regularization to control model complexity.

All three models were trained and evaluated on the **same preprocessed dataset** for a fair, apples-to-apples comparison.

MODEL EVALUATION

Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression ✅	**86.05%**	61.54%	34.04%	43.84%
Random Forest	83.33%	41.67%	10.64%	16.95%
XGBoost	86.05%	65.00%	27.66%	38.81%

Logistic Regression selected as the final model — it achieved the best overall balance across Accuracy, Precision, Recall, F1-Score, and ROC-AUC, with the added benefit of interpretability for HR stakeholders.

DEPLOYMENT

Streamlit Web Application



Application Features

Employee Information Form

Input fields for age, department, job satisfaction, overtime, and more

Real-Time Prediction

Instant attrition risk output powered by the Logistic Regression model

Confidence Score

Displays prediction probability to support informed HR decisions

HR Decision Support

Practical tool for workforce planning and retention strategy

RESULTS & FUTURE SCOPE

What We Built & What's Next

✅ Project Results

- End-to-end ML pipeline from raw data to deployed model
- Interactive Streamlit web application for real-time prediction
- Logistic Regression model achieving **86.05% accuracy**

🔭 Future Scope

SMOTE

Address class imbalance

Hyperparameter Tuning

Optimize model performance

Cross-Validation

Improve generalization

SHAP Explainability

Interpret feature importance

Cloud Deployment

Scale for production use

CONCLUSION

Summary

1

Problem Addressed

Built an ML system to predict employee attrition using the IBM HR Analytics dataset

2

Best Model Selected

Logistic Regression achieved **86.05% accuracy** with the best balance across all evaluation metrics

3

Real-World Deployment

Delivered an interactive Streamlit application enabling real-time, explainable HR predictions

4

Scalable Foundation

Established a reproducible pipeline ready for SMOTE, SHAP, and cloud deployment

---

Thank You

Questions?

Aditya Verma · B.Tech CSE (AI & ML) · IIHMF, MNNIT Prayagraj · Summer 2026